Metastatic Site Classifications of Lung Adenocarcinoma: Comparing Clinical, Genomic, and Combined Feature Sets Across Machine Learning Methods

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1 Introduction

Lung adenocarcinoma is the most common histological subtype of lung cancer and accounts for the majority of cancer-related deaths in the United States. Metastasis is responsible for over 90% of cancer-related mortality [5], and patients without metastases survive approximately three times longer than those with organ metastases [10]. A key biological phenomenon underlying metastatic spread is organotropism, which is the tendency of cancer cells to preferentially spread to specific organs. If metastatic destination can be predicted at the time of primary resection, clinicians could potentially tailor treatment strategies, select targeted therapies, and identify patients for site-specific clinical trials.

In this work, we study whether metastatic sites can be predicted from clinical and genomic features available after time of surgery. We consider both the binary prediction task as a baseline and a more challenging problem in multi-label prediction. This task presents several challenges including small sample size for each metastatic size relative to feature dimensionality, severe class imbalance between the metastatic sites, and the need to model dependencies between the sites.

We evaluated multiple linear and nonlinear machine learning models across across clinical, genomic, combined, and interaction feature sets. We test three core hypothesis: (1) clinical features provide stronger predictive performance than genomic features at this sample size; (2) combining feature sets improves predictive performance; and (3) nonlinear models outperform linear models by capturing complex feature interactions. We also conduct a fairness audit across sex and race subgroups to assess whether model performance varies meaningfully across demographic groups. We find that clinical features consistently outperform genomic features for both binary classification and multilabel classification. Combining feature sets does not meaningfuly improve generalization in the multilabel problem, but does in the binary prediction problem. Furthermore, linear models outperform nonlinear models, which tends to overfit due to limited amount of data. Lastly, we found no substantial sex disparity but insufficient representation of non-White patients to evaluate racial fairness.

2 Background

Most prior ML work in oncology focuses on binary survival prediction [4]. The TCGA 2014 comprehensive molecular profiling of LUAD established key driver mutations including EGFR, KRAS, TP53, STK11, and KEAP1 [9], which actively shape tumor composition and metastatic behavior [3]. Machine learning approaches to cancer prediction are well established [7], with penalized regression being the most popular due to high-dimensional genomic data. Lasso's L1 penalty causes sparsity through implicit feature selection, while Ridge's L2 penalty shrinks coefficients without eliminating them. LDA finds maximally separating linear combinations of features but assumes Gaussian class-conditionals, an assumption violated by binary mutation indicators (our genomic

dataset). Neural networks can capture nonlinear interactions but require substantially more data to generalize reliably on small biomedical datasets [8].

3 Data

3.1 Dataset

Data is sourced from the Lung Adenocarcinoma Met Organotropism dataset (MSK, Cancer Cell 2023) via cBioPortal [1, 2, 5]. Of 2,298 total patients, we restricted analysis to Groups 1 and 2 to ensure clean temporal ordering between feature measurement and outcome: Group 1 contains 262 non-metastatic patients with at least two years of follow-up (57.6%), and Group 2 contains 193 early-stage patients who subsequently developed metastasis (42.4%), giving us 455 samples. Features are measured at primary resection before any metastatic outcome is observed.

The clinical feature set contains 20 features including age, sex, race, tumor stage, and smoking history. The genomic feature set contains 50 features: 38 binary mutation indicators across a LUAD-relevant gene panel, plus 12 continuous summary features including TMB, FGA, ploidy, purity, and MSI score. The combined feature set merges both into a 69-feature matrix, and the combined with interaction set adds 12 biologically motivated interaction terms (e.g., TP53×EGFR, KEAP1×STK11, smoking ×KRAS) for a total of 81 features.

3.2 Label Construction

For binary prediction, $\mathbf{y} \in \{0, 1\}$ indicates ever-metastatic status (42.4% positive). For multi-label prediction, each patient is assigned $\mathbf{y} \in \{0, 1\}^7$ across the seven metastatic sites. Class distribution is severely imbalanced, with Lung (24.0%) and LN (20.9%) being the most frequent, while Adrenal (4.8%, n=22) and Liver (5.1%, n=23) have critically few positives, presenting a fundamental challenge for per-label prediction. (Appendix B, Figure 5).

3.3 Preprocessing

Features with more than 20% missing values were dropped and rows with remaining missing values were removed. All post-sample columns were explicitly excluded to prevent leakage, including treatment history, survival outcomes, and direct metastatic status indicators. The genomic gene panel was refined from 51 to 38 genes: non-LUAD specific oncogenes were removed (e.g. CCND1, CDK4, IDH1, IDH2), while clinically significant low-frequency LUAD drivers were retained (e.g. EGFR, ALK, ROS1, RET). All features were standardized using `StandardScaler` fit on training data only. All datasets used a consistent 80/20 train/test split with `random_state=42`, yielding 364 training and 91 test samples.

4 Methods

4.1 Task Formulation and ClassifierChain

We evaluated two prediction tasks. The primary task is multi-label classification of seven metastatic sites, where each patient may have multiple positive site labels simultaneously. The secondary task is binary classification of ever-metastatic status, serving as a baseline for comparison. Since metastatic sites co-occur frequently (e.g., all patients in the data who had CNS metastasis also had pleura metastasis), all sklearn models are wrapped in `ClassifierChain`, which trains classifiers

sequentially, augmenting each classifier's input with predictions of all preceding label (Appendix B, Figure 6)(Appendix C).

Chain order is CNS → Bone → Liver → Adrenal → Lung → Pleura → LN, placing the rarest and most genomically distinctive sites first to minimize error propagation. CNS had a well-established correlate in EGFR mutation status; LN is placed last as it is least genomically specific.

4.2 Models

All sklearn models used `class_weight='balanced'` to address class imbalance.

Linear Predictors For each site k:

$$z_i^{(k)} = \beta_0^{(k)} + \beta_1^{(k)} x_{i,\text{EGFR}} + \beta_2^{(k)} x_{i,\text{KRAS}} + \cdots + \beta_p^{(k)} x_{i,p} \tag{1}$$

$\beta_j^{(k)}$ is the learned coefficient for feature j at site k. All linear models learn a linear decision boundary of this form. For binary prediction of ever-metastatic status, linear models are trained without `ClassifierChain` wrapper, dropping the site index k.

Logistic Regression

$$\mathcal{L}(\mathbf{w}) = -\sum_{i=1}^{n} [y_i \log \hat{\pi}_i + (1 - y_i) \log(1 - \hat{\pi}_i)] + \frac{1}{C} \|\mathbf{w}\|_p^p \tag{2}$$

$$\pi_i = \frac{1}{1 + e^{-z_i^{(k)}}} \tag{3}$$

where p = 1 (Lasso), p = 2 (Ridge), and C → ∞ (No regularization). Both Lasso and Ridge use GridSearchCV over $C \in \{0.001, 0.01, 0.1, 1, 10, 100, 1000\}$. Lasso uses the `saga` solver with `max_iter=5000` for numerical stability on rare binary labels within the chain.

LDA. Models each class as multivariate Gaussian with shared covariance Σ, classifying by:

$$\delta_k(\mathbf{x}) = \mathbf{x}^\top \Sigma^{-1} \boldsymbol{\mu}_k - \tfrac{1}{2} \boldsymbol{\mu}_k^\top \Sigma^{-1} \boldsymbol{\mu}_k + \log \pi_k \tag{4}$$

Ledoit-Wolf shrinkage regularizes Σ. Expected to underperform on genomic features since binary mutation indicators violate the Gaussian assumption.

SVM. Linear kernel, C tuned via GridSearchCV over $\{0.001, 0.1, 1, 10, 100\}$ (clinical) or $\{0.001, 0.01, 0.1, 1, 10, 100, 1000\}$ (genomic/combined). Linear kernel appropriate for high-dimensional sparse binary features.

MLP. Two hidden layers (64, 32), BatchNorm, ReLU, dropout 0.5/0.4. Trained 200 epochs, Adam lr=10^{-3}, weight decay 10^{-2}. `BCEWithLogitsLoss` with per-label positive weights:

$$w_k = \frac{n_{\text{neg},k}}{n_{\text{pos},k}} \tag{5}$$

Learns nonlinear decision boundaries. Seeds fixed (`torch.manual_seed(42)`, `numpy.random.seed(42)`) for reproducibility.

4.3 Evaluation

Primary metric is macro-average F1:

$$F1_{\text{macro}} = \frac{1}{K}\sum_{k=1}^{K}\frac{2P_kR_k}{P_k+R_k}, \quad P_k = \frac{TP_k}{TP_k+FP_k}, \quad R_k = \frac{TP_k}{TP_k+FN_k} \tag{6}$$

Macro F1 treats all seven labels equally, preventing the Non-Metastatic majority from dominating. GridSearchCV with `cv=5` is applied exclusively to training data, so the 91 test samples are never used during model selection. AUROC was calculated for all four feature sets, but was not for the multi-label task as per-label AUC is unreliable for sites few than 10 test positives (Adrenal: 5, Liver: 6).

4.4 Fairness Audit

Ridge is evaluated across demographic subgroups for all three feature sets. Subgroups with fewer than 10 test samples are excluded. Per-label and macro F1 are reported alongside the gap from overall performance.

5 Implementation

The pipeline is implemented in Python using scikit-learn and PyTorch. Scripts are modularized by stage: `build_genomic_features.py` constructs the genomic matrix; `build_combined.py` merges feature sets; `train.py` and `evaluate.py` train and evaluate sklearn models; `genomic_mlp.py` trains the MLP; `fairness_audit.py` runs demographic subgroup analysis; `visualize.py` generates all figures. Full source code is available at: `[https://github.com/dlee922/cs6140-final-project-jc-dl]`.

6 Results

6.1 Binary Prediction of Metastasis

All results are on the held-out test set of 91 samples. We evaluated binary prediction of ever-metastatic status for all four feature sets. Figure 1 shows AUROC curves for all feature sets. While we reported AUC for completeness, the primary metric for "best model" are based on macro-F1. For clinical features, Logistic Ridge achieved the best results(test F1 = 0.696, AUC = 0.78). (Figure 1A). Notably, all models showed minimal overfitting. Genomic features performed worse, with the best model being Random Forest (test F1= 0.631, AUC = 0.64), with severe overfitting (train F1 = 0.980). The combined feature set improved the performance compared to clinical and genomic features, with better test F1 scores across all models, with the best model being logistic ridge (test F1 = 0.723, AUC = 0.79) Additional interaction terms added to combined feature set further improves SVM test F1 from 0.681 to 0.735 and Random Forest AUC from 0.79 to 0.81. However, adding interactions terms increased overfitting across all models.

6.2 Multi-label Performance Across Feature Sets

6.2.1 Test Macro-F1 Comparison

We used the same models to evaluate its performance on multi-label classification across the four feature sets. Across feature sets, clinical features provided the strongest predictive performance,

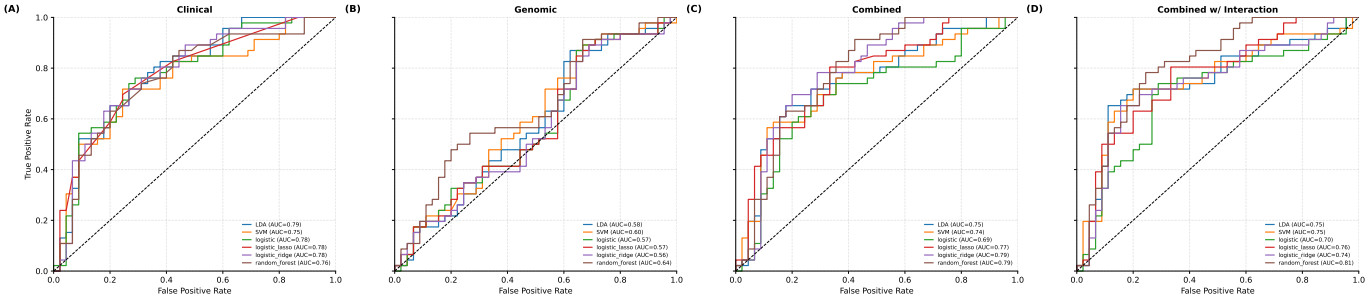

Figure 1: AUROC curves for clinical, genomic, combined, and combined with interaction feature sets. AUC values are macro-averaged across labels. LDA achieves the highest clinical AUC (0.79); genomic models cluster lower, consistent with weaker discriminative signal.

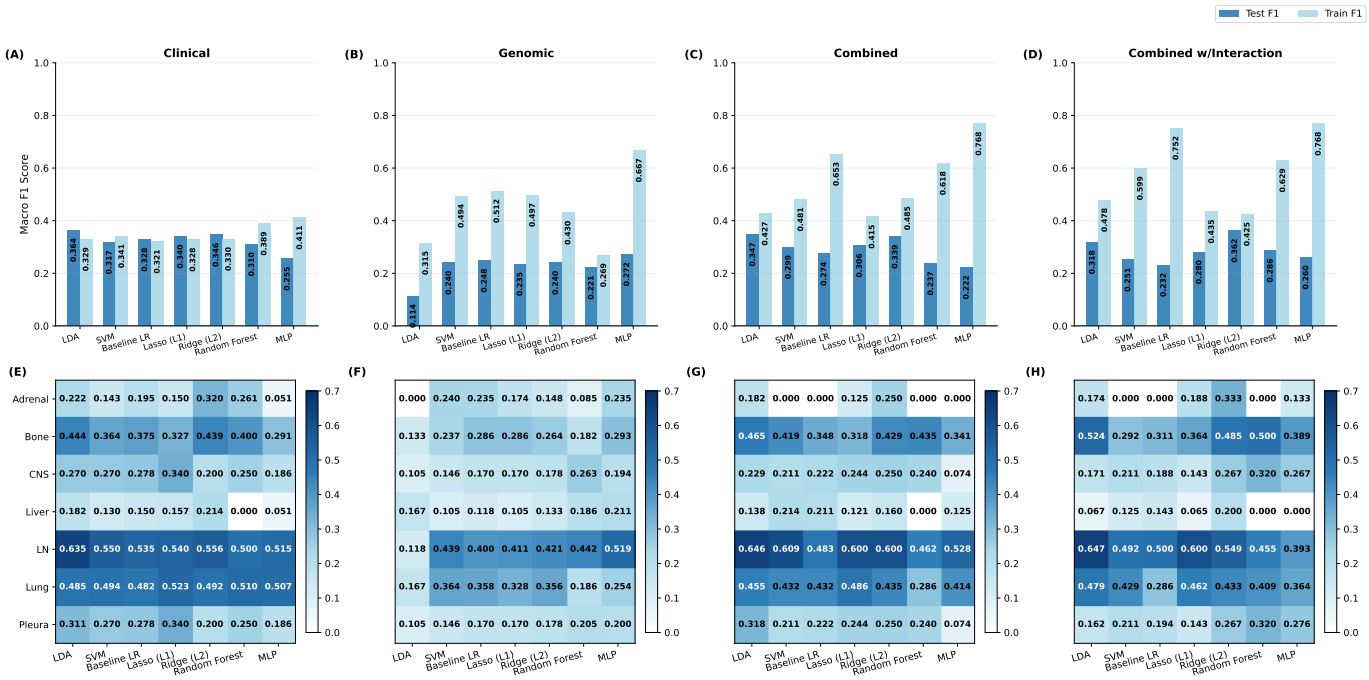

Figure 2: Model performance comparison across four feature sets. Panels A–D show test F1 (dark) and train F1 (light) per model; the gap between bars indicates overfitting. Panel E–H show per-label F1 heatmaps for the corresponding feature set.

with LDA achieving the highest test F1 of 0.364. Genomic features performed substantially worse than clinical, with Baseline Logistic achieving the highest test F1 of 0.248. Combining clinical and genomic features does not meaningfully improve test F1 over clinical features alone. LDA achieves the highest test F1 of 0.347, marginally below the clinical-only LDA of 0.364 (Figure 2C). Adding 12 biologically motivated interaction terms (e.g., TP53×EGFR, smoking×KRAS) to the combined feature set does not improve performance (Figure 2D). Ridge achieves the highest test F1 of 0.362, comparable to clinical-only LDA (0.364).

6.2.2 Linear vs Non-Linear Models

Across all feature sets, linear models generally outperformed nonlinear approaches and exhibited less overfitting. For clinical features, LDA (test F1 = 0.364, train F1 = 0.329), and Ridge (test F1 = 0.346, train F1 = 0.330) showed negative train-test gaps, indicating that these models capture modest but genuine signal without the complexity for memorization. Using genomic features, LDA performed the worst at 0.114, consistent with the Gaussian assumption being violated by binary mutation indicators. In contrast, other linear models such as logistic regression and SVM are better suited to sparse binary features, and thus achieved higher performance compared to LDA. Furthermore, linear models such as Logistic Ridge maintained more controlled overfitting even as dimensionality increases in the combined feature set due to L2 penalty (test F1 = 0.339, train F1 = 0.485)(Figure 2C). In contrast, the overfitting problem is more severe in non-linear models. For example, using clinical features, Random Forest (test 0.310, train 0.389, gap +0.079) and MLP (test 0.255, train 0.411, gap +0.156) showed moderate overfitting. This problem is consistent in the remaining three feature sets. Notably, overfitting is extreme in MLP. For example, using genomic features, MLP achieves test F1 of 0.272 but with the largest overfitting gap (train 0.667, gap +0.395). The gap worsened in the combined feature set, with MLP (test 0.222, train 0.768, gap +0.546). These results demonstrate that 364 training samples are insufficient for a neural network to learn generalizable representations and expanded feature space exacerbates memorization without improving generalization.

6.2.3 Effects of Combining Features and Adding Interaction

Combining clinical and genomic features failed to improve model generalization in any substantial way. The main factors that might explain the models failure to improve include overlapping signals between clinical and genomic features (KRAS mutation rate is 34% in smokers versus 6% in never-smokers [6]), the increase in dimensionality when expanding from 20 to 69 features with only 455 samples, and the genomic panel reflecting general LUAD biology rather than organotropism-specific signals. The interaction terms increased overfitting further: Baseline LR (test 0.231, train 0.752, gap +0.521) and MLP (test 0.260, train 0.768, gap +0.508) show severe memorization. The 81 features relative to 364 training samples pushed the models further into the high-dimensional regime, confirming that feature engineering without additional samples does not improve generalization. regularization.

6.3 Per-Label Analysis

LN and Lung achieved the highest per-label F1 across all models and feature sets (Figure 2E–H), with LN reaching 0.647 (combined with interaction, LDA) and Lung reaching 0.523 (clinical, Lasso). Liver and Adrenal performed consistently poorly across all conditions (Liver 0.000–0.214, Adrenal 0.000–0.333). With only 5–6 positive tests respectively, these scores are unreliable and reflect a data

limitation rather than a modeling failure. CNS showed consistent signal across feature sets (0.105–0.340), consistent with the established EGFR–brain tropism relationship [9]. Bone performed moderately well (0.133–0.524), with stronger signal in the combined and interaction feature sets.

6.4 Fairness Audit

No meaningful sex disparity is detected across any pipeline. The largest sex gap occurs in the clinical pipeline (Male: 0.397, Female: 0.325, gap +0.072) and the smallest in the combined pipeline (Male: 0.357, Female: 0.341, gap +0.016). For race, only White patients ($n = 76$, 84.8%) have sufficient test samples. Performance gaps for White patients are negligible (−0.012 to +0.012). Black and Other subgroups have 1 and 3 test samples respectively and cannot be evaluated, meaning this is a dataset limitation that would require multi-institutional data to address.

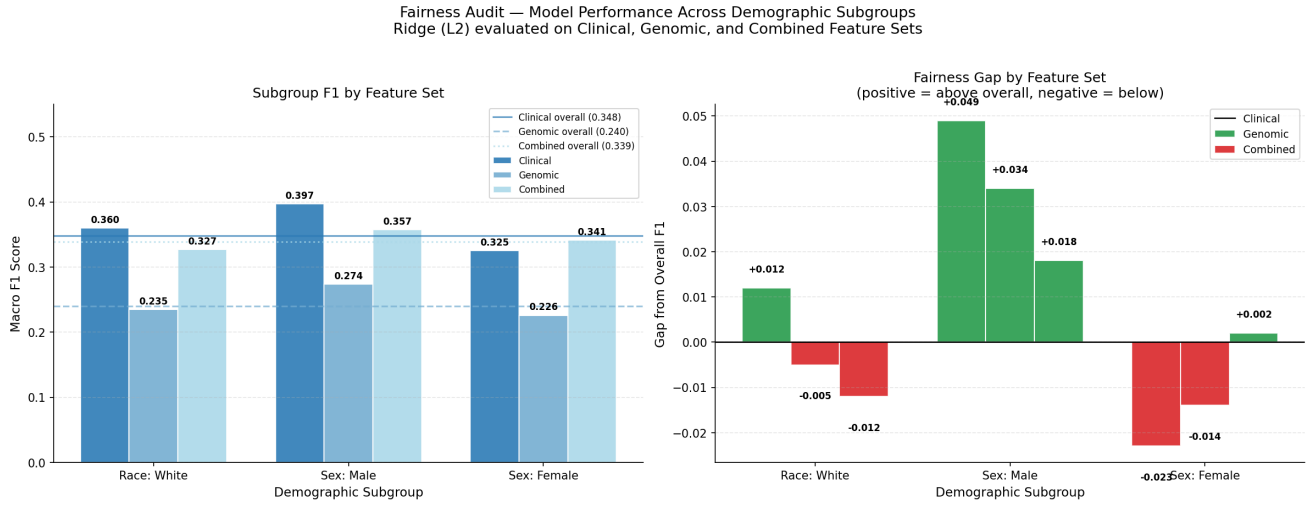

Figure 3: Fairness audit for Ridge (L2) across clinical, genomic, and combined feature sets. Left: absolute macro F1 per demographic subgroup with overall performance reference lines. Right: gap from overall F1 (positive = above overall, negative = below). No meaningful sex disparity is observed; racial subgroup analysis is limited to White patients due to insufficient sample sizes for other groups.

7 Discussion

Three main findings emerged from this project. First, clinical features provided stronger predictive signal than genomic features at this sample size. Despite the Gaussian assumption, LDA performed well on clinical features likely because more features are continuous rather than binary, and standardization may have helped the data approximate a Gaussian distribution. By contrast, LDA on genomic features performed worst across all models and feature sets, highlighting the importance of matching model assumptions to data structure. While organotropism has a biological basis in tumor genomics [5], the signal-to-noise ratio in the binary mutation matrix is low at 455 samples. This is consistent with Lengel et al., 2023, where proportions of EGFR and KRAS mutations were similar between metastatic and non-metastatic patients. Features such as tumor stage and smoking history provide cleaner aggregate signal than sparse mutation profiles.

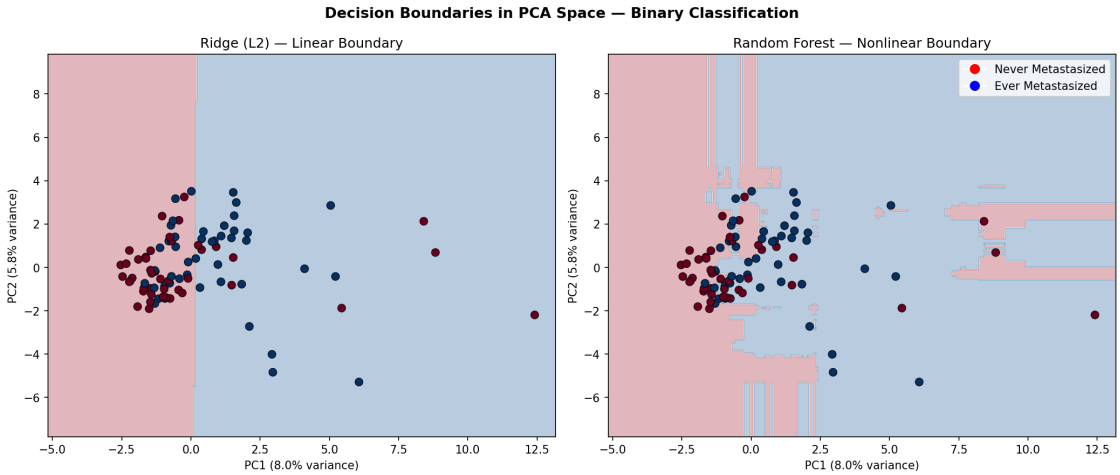

Figure 4: Decision boundaries in PCA space for binary classification (metastasized vs. not). Ridge (L2) learns a linear boundary while Random Forest produces a nonlinear boundary that overfits to training noise. The substantial overlap between classes in the first two principal components (explaining 13.8% of variance) illustrates the inherent difficulty of the prediction task.

Second, combining feature sets did not improve generalization in the multi-label task. This is consistent with overlapping information content between clinical and genomic streams (e.g., KRAS mutation rate is 34% in smokers versus 6% in never-smokers [6]) and the curse of dimensionality at this sample size. However, for binary prediction, interaction terms did improve most linear models, likely due to the relatively balanced label distribution. In the multi-label task, the interaction terms do not capture sufficient independent signal to offset increased dimensionality, suggesting that feature interactions help generalization primarily when sample size is adequate and classes are balanced.

Third, linear models consistently outperformed nonlinear approaches. The decision boundary visualization (Figure 4) illustrates this contrast: Ridge learned a clean linear boundary in PCA space while Random Forest produced a fragmented nonlinear boundary that captures training noise rather than generalizable structure. Furthermore, with only 364 training samples distributed across 7 binary output labels, neural networks lack sufficient data for generalization, consistent with the broader literature on small biomedical datasets where linear inductive bias provides a regularization advantage [7].

Limitations. The 455-sample dataset is small relative to the task complexity. Per-label F1 for Liver and Adrenal is unreliable with only 5–6 test positives. The MSK cohort's demographic composition (84.8% White) prevents fairness evaluation for non-White patients. The genomic panel was curated for general LUAD biology rather than organotropism prediction specifically.

Future work. Multi-institutional data aggregation would simultaneously improve performance and enable racial fairness evaluation. A targeted organotropism gene panel with clinical-genomic interaction features could improve genomic signal contribution. Integration of survival analysis and exploration of dedicated multi-label architectures represent additional directions.

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A Supplementary Tables

Tables 1–3 provide numerical results corresponding to Figure 2 panels A–C. Gap is computed as Train F1 minus Test F1; negative values indicate the model generalizes slightly better than its training performance suggests.

Table 1: Clinical feature results, sorted by test F1.

Model	**Test F1**	**Train F1**	**Gap**
LDA	0.364	0.329	−0.035
Ridge (L2)	0.346	0.330	−0.016
Lasso (L1)	0.340	0.328	−0.012
Baseline LR	0.328	0.321	−0.007
SVM	0.317	0.341	+0.024
Random Forest	0.310	0.389	+0.079
MLP	0.255	0.411	+0.156

Table 2: Genomic feature results, sorted by test F1.

Model	Test F1	Train F1	Gap
LDA	0.114	0.315	+0.201
Ridge (L2)	0.240	0.430	+0.190
Lasso (L1)	0.235	0.497	+0.262
Baseline LR	0.248	0.512	+0.264
SVM	0.240	0.494	+0.254
Random Forest	0.221	0.269	+0.048
MLP	0.272	0.667	+0.395

Table 3: Combined feature results, sorted by test F1.

Model	Test F1	Train F1	Gap
LDA	0.347	0.427	+0.080
Ridge (L2)	0.339	0.485	+0.146
Lasso (L1)	0.306	0.415	+0.109
SVM	0.299	0.481	+0.182
Baseline LR	0.274	0.653	+0.379
Random Forest	0.237	0.618	+0.381
MLP	0.222	0.768	+0.546

B Supplementary Figures

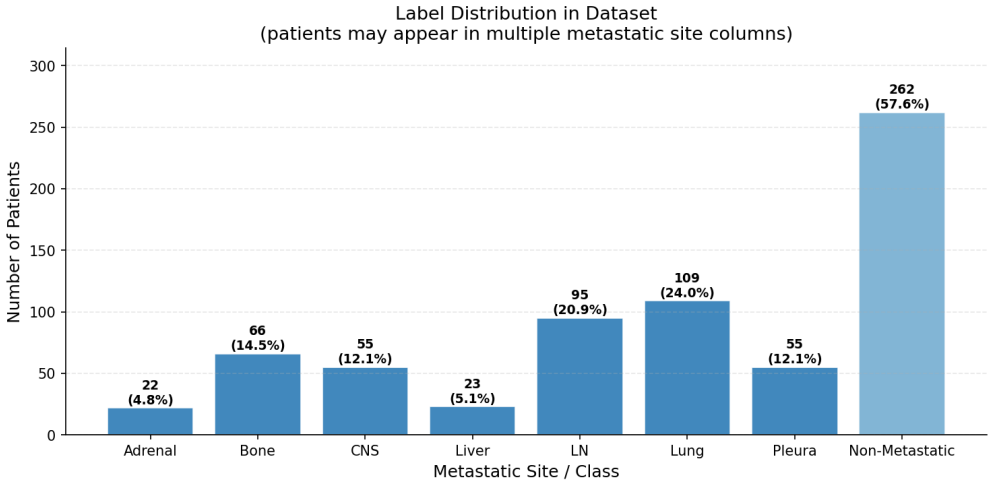

Figure 5: Distribution of metastatic site labels. Non-metastatic class is the majority at 57.6%, followed by lung (24.0%) and lymph node (20.9%).

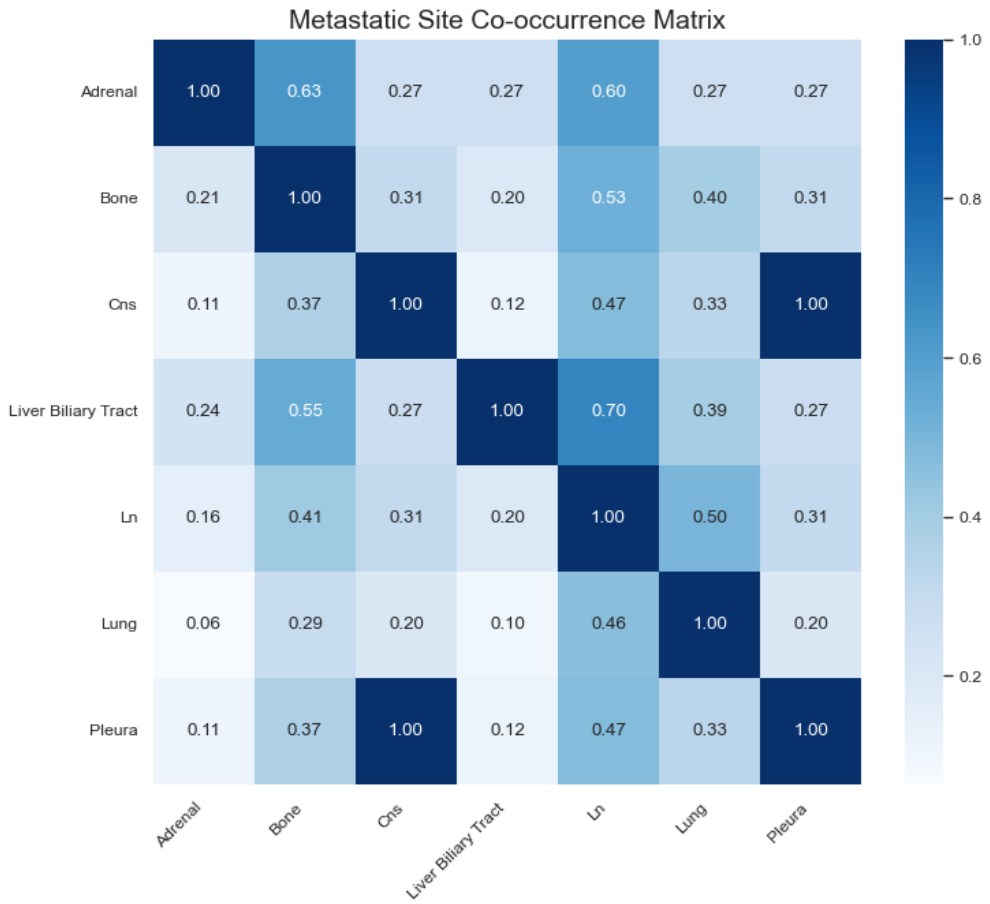

Figure 6: Co-occurrence matrix of metastatic site labels. Each cell shows the proportion of patients positive for the row label who are also positive for the column label. High co-occurrence between sites such as LN and Lung motivates the multi-label formulation over single-label multiclass classification.

C ClassifierChain Formulation

For the multi-label task, each label k in the chain is predicted as:

$$\hat{y}_k = f_k(\mathbf{x}, \hat{y}_1, \ldots, \hat{y}_{k-1}) \tag{7}$$

Data and Code Availability

The MSK Cancer Cell 2023 Met Organotropism dataset is publicly available via cBioPortal at `https://www.cbioportal.org`.

Full source code is available at: `[https://github.com/dlee922/cs6140-final-project-jc-dl]`.

Statement of Contributions

Jason Chen: Initial Data Exploration, Training of Clinical/Combined Feature Sets, Code Structure

Daniel Lee: Mutation Matrix Generation, Training Genomic Feature Set, Fairness Audit, MLP Implementation, Visualizations